Q1.

1.

```
use "C:\Users\Lenovo\Desktop\C...I\eco3121\ass3\aghousehold(1).dta", clear

keep if year==2001

display "Q1"
display "1."

gen yield = hx9S_nh12S / hx9S_nh112
gen fertilizer = nh6_3 / hx9S_nh112
gen migration = (h9S_nh269 > 0)
gen log_yield = log(yield + 1)
gen log_fertilizer = log(fertilizer + 1)
```

2.

```
. regress migration log_yield
```

Source	SS	df	MS	Number of obs	=	4,200
Model	3.68868145	1	3.68868145	F(1, 4198)	=	14.88
Residual	1040.44251	4,198	.247842427	Prob > F	=	0.0001
				R-squared	=	0.0035
				Adj R-squared	=	0.0033
Total	1044.13119	4,199	.24866187	Root MSE	=	.49784

migration	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
log_yield	.0495398	.0128412	3.86	0.000	.0243642	.0747155
_cons	.1773844	.0743338	2.39	0.017	.0316509	.3231179

Interpretation: The coefficient of log(yield) is 0.049s398,

It means that holding other factors constant, on average 1% increase in yield would bring 0.049s% increase in the probability of migration.

The p-value of the coefficient is $0.000 < 0.01$

∴ It is statistically significant.

3. ① Omitted variable bias:

There may be variables affecting both yield and migration. The direction of bias would be ambiguous.

For example, the household education level (educ)

∵ Cov(educ, yield) > 0 according to the common sense,

Cov(migration, yield) > 0 according to the regression model.

∴ There would be an upward bias.

② Measurement error:

The yield variable may be measured with error due to wrong measuring or wrong reporting.

There may be a downward bias because the effect of yield may be underestimated.

③ Simultaneity (Reverse causality)

We know that yield would affect migration. In the meantime, yield is also affected by migration.

If migration leads to lower yields (consistent with common sense), the coefficient can't accurately reflect the true effect. Then the bias direction would be downward

If migration leads to higher yields (against common sense but possible), the coefficient can't accurately reflect the true effect. Then the bias direction would be upward

4. The first stage regression model.

$log_yield_i = \pi_0 + \pi_1 log_av_rain_i + v_i$

```
. regress log_yield log_av_rain
```

Source	SS	df	MS	Number of obs	=	4,003
Model	182.285667	1	182.285667	F(1, 4001)	=	560.65
Residual	1300.86357	4,001	.325134609	Prob > F	=	0.0000
				R-squared	=	0.1229
				Adj R-squared	=	0.1227
Total	1483.14924	4,002	.370602008	Root MSE	=	.57021

log_yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
log_av_rain	.4644991	.0196173	23.68	0.000	.4260382	.50296
_cons	2.526537	.1365663	18.50	0.000	2.258791	2.794284

Interpretation: The coefficient of log_av_rain is 0.4645

It means that holding other factors constant, on average 1% increase in average rainfall would increase the yield by 0.4645%

$R^2 = 0.1229$, it means that 12.29% variation in log_yield is explained by log_av_rain.

The p-value of the coefficient is $0.000 < 0.01$, ∴ It is statistically significant.

5. The second stage regression model:

$migration_i = \beta_0 + \beta_1 \widehat{log_yield}_i + u_i$

```
. regress migration log_yield_hat
```

Source	SS	df	MS	Number of obs	=	4,003
Model	33.6398682	1	33.6398682	F(1, 4001)	=	140.16
Residual	960.267701	4,001	.240006924	Prob > F	=	0.0000
				R-squared	=	0.0338
				Adj R-squared	=	0.0336
Total	993.907569	4,002	.248352716	Root MSE	=	.48991

migration	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
log_yield_hat	.4295868	.0362857	11.84	0.000	.3584465	.500727
_cons	-2.012801	.208899	-9.64	0.000	-2.422359	-1.603243

Interpretation: The coefficient of log_yield_hat is 0.4296

It means that holding other factors constant, on average 1% increase in predicted yield would increase the probability of migration by 0.4296%.

$R^2 = 0.338$, it means that 338% of variation in migration is explained by the instrumented yield variable.

The p-value of the coefficient is 0.000 < 0.01 ∴ It is statistically significant.

6.

```
. ivregress 2sls migration (log_yield = log_av_rain)

Instrumental variables 2SLS regression          Number of obs   =      4,003
                                                Wald chi2(1)    =     111.36
                                                Prob > chi2     =     0.0000
                                                R-squared       =          .
                                                Root MSE        =     .54961
```

migration	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
log_yield	.4295868	.0407081	10.55	0.000	.3498003	.5093732
_cons	-2.012801	.234359	-8.59	0.000	-2.472136	-1.553466

Endogenous: log_yield
Exogenous: log_av_rain

(a) Compared with the model in (5):

The coefficients of the two models are the same.

The standard errors of the coefficients using ivregress 2sls command is larger.

(b) Comparison:

The coefficient of log_yield in LPM is 0.049s. with the standard error of 0.0128

$\therefore \hat{\beta}_{1,IV} > \hat{\beta}_{1,LPM}$, there is the problem of endogeneity.

It means that there is a downward bias, maybe caused by measurement error or reverse causality or omitted variables.

(∴ It meets our expectation.

Q2. $Crop = \alpha_0 + \beta_0 Fertilizer + u$

$Fertilizer = \alpha_1 + \beta_1 Crop + v$

1. $\beta_0 > 0$: According to our common sense, within a reasonable range, fertilization is beneficial to crops.
So we expect that more fertilizer usage would increase the crop yield.
$\therefore \beta_0 > 0$

$\beta_1 > 0$: Because as crop yields increase, farmers' demand for fertilizers also increases.
So higher crop yields mean that farmers may need more fertilizers
$\therefore \beta_1 > 0$

2. The OLS estimator for β_0 and β_1 are biased because of simultaneity (reverse causality)
In the first equation, fertilizer affects crop.
But in the second equation, crop also affects fertilizer.
That brings endogeneity due to simultaneity.

Because fertilizer and crop are positively correlated,

higher fertilizer/crop would bring higher crop/fertilizer.

$\therefore Cov(Fertilizer, u) > 0$, $Cov(Crop, v) > 0$

$\therefore$ For the direction of bias in β_0, β_1, both of them would be upward.

3. We would choose Sunshine as the instrument variable for Crop:
Relevance: Sunshine is correlated with Crop, as growth of crop relies on sun
Exogeneity: Sunshine is only determined by the natural, not affected by other factors.
And sunshine wouldn't affect fertilizer and other factors, uncorrelated with the error term.
$\therefore Cov(Sunshine, Crop) \neq 0$, $Cov(Sunshine, v) = 0$

We would choose Budget as the instrument variable for Fertilizer.
Relevance: Farmer's budget affects the usage of fertilizer.
$\therefore$ Usage of fertilizer is correlated with budget.
Exogeneity: Budget wouldn't directly affect fertilizer and other factors.
It is not determined by crop and other factors neither.
$\therefore Cov(Budget, Fertilizer) \neq 0$, $Cov(Budget, u) = 0$

$Fertilizer_i = \pi_0 + \pi_1 Budget_i + a_i$

$Crop_i = \alpha_0 + \alpha_1 \widehat{Fertilizer_i} + b_i$

$Crop_i = \delta_0 + \delta_1 Sunshine_i + c_i$

$Fertilizer = \beta_0 + \beta_1 \widehat{Crop_i} + d_i$

q1.do

```
use "C:\Users\Lenovo\Desktop\大二上\eco3121\ass3\aghousehold(1).dta", clear

keep if year==2001

display "Q1"
display "1."

gen yield = hx95_nh125 / hx95_nh112
gen fertilizer = nh6_3 / hx95_nh112
gen migration = (h95_nh269 > 0)
gen log_yield = log(yield + 1)
gen log_fertilizer = log(fertilizer + 1)

display "2."

regress migration log_yield

display "4."

merge m:1 vl_id using "C:\Users\Lenovo\Desktop\大二上\eco3121\ass3\village_rainfall.dta"
tab _merge

gen log_av_rain = log(av_rain)
drop if missing(log_av_rain, log_yield, migration)
regress log_yield log_av_rain

display "5."

predict log_yield_hat, xb
regress migration log_yield_hat

display "6."

ivregress 2sls migration (log_yield = log_av_rain)
```